SUPERSEDING
MIL-W-26217A (USAF)
13 June 1958

MILITARY SPECIFICATION

WINCH, DRUM, POWER OPERATED AF/E24H-1, CARGO LOADING, 4,000 POUND CAPACITY, VARIABLE SPEED, 28V DC, PORTABLE

1. SCOPE

1.1 This specification covers one type of 28 Volt (V) direct current (DC), variable speed, 4,000 pound capacity, cargo loading, power-operated, portable, aircraft-cargo winch, designated AF/E24H-1.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Military

MIL-T-704	Treatment and Painting of Material
MIL-D-1000	Drawings, Engineering and Associated Lists
MIL-C-1511	Wire Rope; Steel (Carbon) Flexible, Preformed
MIL-W-5088	Wiring, Aircraft, Installation of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification For
MIL-S-56/6	Splicing; Cable Terminal, Process For Aircraft
MIL-S-6743	Switches, Sensitive and Push, Snap Action, Actuators and Enclosures, General Specification For
MIL-E-708C	Electric Equipment, Aircraft, Selection and Installation Of
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification For
MIL-A-8421	Air Transportability Requirements, General Specification For
MIL-M-8609	Motors, Direct-Current, 28-Volt System, Aircraft, General Specification For

STANDARDS

Federal

FED-STD-595	Colors

FSC 3950

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:19Z
Check the source to verify that this is the current version before use.

Military

MIL-STD-129	Marking For Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-162	Material Handling Equipment, Self-Propelled: Preparation For Delivery and Storage
MIL-STD-838	Lubrication Of Military Equipment
MS33586	Lubrication Of Military Equipment Metals, Definition of Dissimilar

Air Force-Navy Aeronautical

AN2552	Receptacle - External Power, 28 Volts DC

PUBLICATION

Air Force-Navy Aeronautical Bulletin

143	Specifications and Standards: Use Of

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Components. The winch shall consist of the following components and necessary accessory equipment: Motor, clutch, cable drum and cable assemblies, and overload safety device.

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with Bulletin 143 except as provided in 3.3.1 and 3.3.2.

3.3.1 Commercial parts. Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are not suitable standard parts. In any case, commercial utility parts, such as screws, bolts, nuts, and cotter pins, having suitable properties may be used provided:

a. They can be replaced by the standard parts (MS or AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts list, and if practical, on the contractor's drawings.

3.3.2 Standard parts. With the exceptions in 3.3.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.4 Materials.

3.4.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with eact other. Dissimilar metals are defined in MS33586.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:19Z
Check the source to verify that this is the current version before use.

3.4.2 Protective treatment. When materials are used in the construction of the winch that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.5 Design and construction. The winch shall be designed and constructed so that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.5.1 Adjustments and repairs. The winch shall be constructed so that adjustments, repairs, and minor overhauls can be accomplished by personnel of operating units and overhaul bases with the use of standard hand tools.

3.5.2 Design. The winch shall be designed as a portable, self-contained unit, except for the external power source. The winch assembly, exclusive of the cable, shall be designed structurally using a safety factor of 3 based on the winch rated capacity. The base of the winch shall be designed to support the winch during cargo loading operations without damaging the floor of cargo aircraft. Rollers shall be built into the base to facilitate moving the winch. All rollers shall be at least 4-inches long and faced with a semisoft rubber compound. Handles shall be provided as required to facilitate moving the winch.

3.5.3 Life. The winch shall be designed for a life expectancy of 2,000 hours without any overhaul other than periodic servicing and adjustment, except that brushes may be replaced at the end of 1,000 hours. Periodic inspection and servicing shall not be required at intervals of less than 200 hours.

3.5.4 Air transportability. The winch shall be designed and constructed to meet the air transportability requirements of MIL-A-8421.

3.5.5 Hoisting rings. The winch shall be provided with hoisting rings to permit lifting with an overhead hoist. The rings shall be at least 3-inches in diameter.

3.5.6 Tiedown provisions. To permit securing the winch to the floor of aircraft for loading and unloading cargo, the winch shall have provisions for rapidly securing it to two 5,000-pound-capacity tiedown rings having minimum clear openings of 3/4-inch diameter, maximum cross-section diameters of 7/8-inch, and arranged in a 20-inch grid.

3.6 Performance. The winch shall be capable of satisfactorily withstanding or operating under the following conditions, as applicable:

a. Temperatures ranging from -65 degrees (^{o}) to plus (+) 160° Fahrenheit (F).

b. Relative humidity up to 100 percent (%).

c. Exposure to salt-sea atmosphere.

d. Sand and dust particles as encountered in desert areas.

e. Explosive atmosphere as encountered in explosive vapor within and surrounding the equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:19Z
Check the source to verify that this is the current version before use.

3.6.1 Capacity. The winch shall be capable of exerting a force of 4,000 pounds on a single purchase line at any speed from 0 to 36 feet per minute (fpm).

3.6.2 Operating position. The winch shall be capable of reeling the full length of cable in and out when the cable is positioned at angles up to 10° to either side of a vertical plane perpendicular to the cable drum with the projection of the cable within the plane at any angle between 0° and 30° above horizontal.

3.7 Motor. An intermittent-duty-type motor conforming to MIL-M-8609, Class B, having a duty cycle and power capability to meet the requirements stated in 3.6.1 and the test specified in Section 4, shall be furnished. The qualification requirements specified in MIL-M-8609 shall not apply. Radio-noise-suppression measurement shall be made at a distance of 25 feet (ft).

3.7.1 Power consumption. With rated load (4,000-pounds) being reeled in at a rate of 36 fpm, current consumption shall not exceed 240 amperes.

3.8 Clutch. The clutch shall be designed to slip and hold any cable load up to 4,000 pounds or regulate the speed of the cable drum so that it will reel in or play out cable at any speed between 0 and 36 fpm under the load specified herein. When playing out cable under load with power removed from the winch, the clutch shall function as a brake and shall regulate the drag on the cable drum and the cable speed. This braking function shall also enable the operator to hold any load up to 4,000-pounds without power to the winch. The clutch shall also incorporate a dead-man brake capable of holding any load up to 4,000-pounds with or without power applied to the winch. The dead-man brake shall be so designed that when the clutch-control lever is released by the operator, the brake will be automatically actuated.

3.8.1 Clutch control. A single clutch-control lever shall be conveniently located on the winch in such a manner as to facilitate ease of operation. The force required to operate the clutch-control lever with any load up to 4,000-pounds shall not exceed 30 pounds. The clutch control shall be so designed that when the winch operator releases the lever, the dead-man brake specified in 3.8 will be actuated.

3.9 Cable drum and cable assemblies.

3.9.1 Cable-drum assembly. The winch shall incorporate a scored, horizontal-mounted, clutch-operated cable-drum assembly without capstan. The cable drum shall be of such design that the dead-man brake can be released and the cable unreeled from the cable drum without the winch operator holding the clutch-control lever. When unreeling cable in this manner, the tension on the cable drum shall be sufficient to prevent excess cable from unreeling. The cable tension force required to unreel the cable shall not exceed 20 pounds.

3.9.1.1 Cable-drum housing. A housing shall be provided to cover as much of the cable-drum assembly as practical. The housing shall have an opening covered by 1/2-inch mesh so that the operator can view the cable drum to determine if the cable is winding properly.

3.9.2 Cable assembly. The cable assembly shall consist of a hook, a cable thimble, and 100 ft of 9/32-inch 7 by 19 preformed flexible steel cable conforming to MIL-C-1511. The hook shall be designed to engage a ring having a minimum clear opening of 3-inches in diameter and a maximum cross sectional diameter of 1-3/4-inches. The hook shall be attached to the cable assembly by splicing in accordance with MIL-S-5676 or swaging in accordance with the best commercial practice.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:19Z
Check the source to verify that this is the current version before use.

3.10 Overload safety device. An automatic safety device shall be provided to prevent the line pull from exceeding 4,700 ± 400 pounds (lbs). Any mechanism that is subject to rapid deterioration from use or that releases the cable drum when it functions will not be acceptable.

3.11 Electrical wiring and accessories.

3.11.1 Wiring and circuitry. All electrical wiring and components integral with and pertinent to the winch shall be in accordance with MIL-W-5088, MIL-E-7080, and the applicable component specifications. Grounds for the electrical system in the winch shall be through the negative return cable only. The wiring shall be arranged so that under no condition will it be possible for operating current to flow through the winch frame or tow-in cable into the airframe of the aircraft in which the winch is mounted.

3.11.2 Electrical junction box. A watertight electrical junction box shall be mounted on the winch. The junction box shall be provided with an on-off power switch conforming to MIL-S-6743, a male receptacle conforming to AN2552, and all other necessary components.

3.12 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-D-1000 shall govern changes in the manufacturer's part numbers.

3.13 Lubrication. Lubricants used in the winch shall conform to MIL-STD-838 and shall be suitable for use under all operating conditions specified herein.

3.14 Screw threads. Screw threads shall conform to MIL-S-7742.

3.15 Dimensions and weight. The dimensions and weight of the winch shall be held to the minimum consistent with the best commercial and engineering practices covering this type of equipment. The weight of the complete winch shall not exceed 250 pounds.

3.16 Color. The color of the winch shall be gloss yellow conforming to FED-STD-595, color 13538.

3.17 Finish. The winch shall be finished in accordance with MIL-T-704.

3.18 Marking.

3.18.1 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.18.2 Instructions. Brief instructions for operating the winch together with a wiring diagram shall be suitably and permanently mounted on the winch. The instructions and diagram shall be such that dirt, oil, exposure to light, etcetera, will not affect their legibility.

3.19 Workmanship.

3.19.1 General. The winch, including all parts and accessories, shall be fabricated and finished in a workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring, and riveting; alignment of parts and tightness of assembly screws and bolts; etcetera.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:19Z
Check the source to verify that this is the current version before use.

3.19.2 Cleaning. The winch shall be thoroughly cleaned and loose, spattered, or excess solder, metal chips, and other foreign material removed during and after final assembly.

3.19.3 Gear assemblies. Gear assemblies shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots, or any other irregularities.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of tests. The inspection and testing of winches shall be classified as follows:

a. Acceptance tests . . . . . See 4.2

b. Preproduction testing . . . . . See 4.4.

4.2 Acceptance tests. Acceptance tests shall consist of:

a. Individual tests

b. Periodic sampling plan and test.

4.2.1 Individual tests. Each winch shall be subjected to the following tests as described under 4.3:

a. Examination of product

b. Operational load test.

4.2.2 Periodic sampling plan and test. One winch shall be selected at random from the first ten and one from the next fifty or fraction thereof, produced and subjected to the overload safety device test described under 4.3 except that the cycle shall be repeated two times in lieu of five.

4.2.2.1 Rejection and retest. When one winch selected from a production run fails to meet the specification, no winches still on hand or later produced shall be accepted until the extent and cause of failure are determined.

4.2.2.2 Individual test may continue. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of winches on hand or produced later shall not be made until it is determined that they meet all the requirements of the specification.

4.3 Test methods.

4.3.1 Examination of product. The winch shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

4.3.2 Operational load test. The winch shall be subjected to 2 cycles of the test specified in 4.3.5 except that the cable speed shall be 30 fpm.

4.3.3 Overload safety device. With the winch cable securely anchored, the clutch shall be engaged. The overload safety device shall function when the line pull exceeds the limits specified in 3.10. The overload safety device shall be allowed to operate for a minimum of 3 minutes. This cycle shall be repeated five times, stopping 2 minutes between cycles. At the completion of the test, the overload safety device shall be operable as specified herein and there shall be no indication of impending failure.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:19Z
Check the source to verify that this is the current version before use.

4.3.4 Environmental tests. The winch shall be subjected to the following environmental tests in accordance with the specified procedures of MIL-E-5272.

4.3.4.1 Low temperature. The winch shall be subjected to a low-temperature test in accordance with procedure I. At the conclusion of this test and at the specified temperature, the winch shall be subjected to the operational load test specified in 4.3.2. If the test chamber is too small for this test, the test may be conducted at room temperature within 3 minutes after removing the winch from the test chamber.

4.3.4.2 High temperature. The winch shall be subjected to a high-temperature test in accordance with procedure I. At the conclusion of this test and at the specified temperature, the winch shall be subjected to the operational load test specified in 4.3.2. If the test chamber is too small for this test, the test may be conducted at room temperature within 3 minutes after removing the winch from the test chamber.

4.3.4.3 Humidity. The winch shall be subjected to a humidity test in accordance with procedure I. At the completion of the test, the winch shall be subjected to the test specified in 4.3.2.

4.3.4.4 Salt spray. The winch shall be subjected to the salt-spray test specified in procedure I.

4.3.4.5 Sand and dust. At the completion of the salt-spray test, the winch shall be subjected to a sand and dust test in accordance with procedure I. The winch shall then be subjected to the test specified in 4.3.2.

4.3.4.6 Explosion-proof test. The winch shall be subjected to an explosion-proof test in accordance with procedure I.

4.3.5 Partial life test. After completion of the environmental tests, the winch shall be subjected to the following 1,000-cycle test. Malfunction which would affect satisfactory operation as specified herein shall be cause for rejection. The cable speed shall be 10 fpm for the first 400 cycles, 25 fpm for the second 400 cycles, and 36 fpm for the remaining 200 cycles. A cycle shall consist of the following:

a. With rated line pull applied to the winch and using the clutch as a brake, unwind the maximum length of cable stopping every 15 feet for 5 seconds. Stop 1 minute before continuing cycle.

b. With rated line pull applied to the winch, wind in the maximum length of cable stopping every 15 feet for 5 seconds. Stop 2 minutes before starting the next cycle.

4.3.6 Air transportability. The winch shall be subjected to air transportability tests in accordance with MIL-A-8421.

4.3.7 Functional test. The winch shall be tested to demonstrate compliance with 3.6.1, 3.6.2, 3.8, 3.8.1 and 3.9.1. The winch shall be mounted so that it may be positioned to reel cable in and out at angles up to 10° either side of a plane perpendicular to the cable drum axis, and at angles 0° to 30° above horizontal. A dead weight load of 4,000 lbs plus or minus 40 pounds (or the equivalent) shall be placed on the cable and the following operations performed. These operations shall be performed continuously, i.e., not more than one-minute shall lapse between the successive cycles of the individual tests and not more than five-minutes shall lapse between tests where a change in position is stated. For the purpose of these tests, a cycle shall consist of continuous reeling in 90 ft

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:19Z
Check the source to verify that this is the current version before use.

of cable at 36 fpm and the continuous reeling out of 90 ft of cable at 36 fpm $\pm$ 6 fpm. Current consumption during the following tests shall be recorded. Current in excess of 240 amperes shall be cause for rejection:

a. Test Number 1, 25 cycles with cable perpendicular to the horizontal and vertical axes of the drum.

b. Test Number 2, 25 cycles with cable 10° to the left side of vertical plane perpendicular to the drum.

c. Test Number 3, 25 cycles with cable 10° to the right side of a vertical plane perpendicular to the drum.

d. Test Number 4, 25 cycles with cable 10° to the left side of a vertical plane perpendicular to the drum and 30° above a line horizontal to the winch.

e. Test Number 5, 25 cycles with cable 10° to the right side of a vertical plane perpendicular to the drum and 30° above a line horizontal to the winch.

4.3.8 Final disassembly inspection. At the completion of the preproduction tests, the winch shall be disassembled and all parts thoroughly examined for distortion, excessive wear, failure, and other unsatisfactory conditions.

4.3.9 Motor test. The motor supplied as a component of the winch shall successfully pass the following tests specified in MIL-M-8609: Examination of Product; Load test (intermittent-duty motors); Dielectric strength; Overspeed; Extreme temperature operation; Radio interference; Operating position; Starting; Life; Humidity; Explosion-proof; and the Performance tests except that calibration and efficiency are not required.

4.3.10 Power consumption test. During the tests specified in paragraphs 4.3.5 and 4.3.7, current consumption shall not exceed 240 amperes. For the purpose of this test, transients of short duration which occur at time of starting shall not be considered.

4.4 Preproduction testing.

4.4.1 Preproduction test sample. The preproduction test sample shall consist of one winch representative of the production equipment. It shall be tested at a laboratory approved by the procuring activity or, when so specified in the contract, at the contractor's plant under the supervision of the procuring activity.

4.4.2 Preproduction tests. Preproduction tests shall consist of all the tests specified under 4.3.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A. The winch shall be preserved and packaged in accordance with MIL-STD-162.

5.1.2 Level C. The winch shall be preserved and packaged in accordance with the manufacturer's commercial practice.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:19Z
Check the source to verify that this is the current version before use.

5.2 Packing.

5.2.1 Levels A and B. Winches preserved and packaged as specified in 5.1.1 shall be packed in accordance with MIL-STD-162.

5.2.2 Level C. Packages which require overpacking for acceptance by the carrier shall be packed in exterior-type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall meet Consolidated Freight Classification Rules or regulations of other common carriers as applicable to the mode of transportation.

5.3 Marking of shipments. Interior and exterior shipping containers shall be marked in accordance with MIL-STD-129. The shipment marking nomenclature shall be as follows:

Winch, Drum, Power Operated AF/E24H-1, Cargo Loading, 4,000-Pound Capacity, Variable Speed, 28V DC, Portable.

6. NOTES

6.1 Intended use. The AF/E24H-1 winch covered by this specification is intended for use in loading and unloading cargo in and out of aircraft.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Selection of applicable levels of preservation, packaging, and packing (see Section 5).

6.2.1 Preproduction test sample. One winch will be required as a preproduction sample to be subjected to the preproduction tests to determine compliance with the requirements of this specification. The point of inspection for these tests will be specified.

Custodians:

Air Force - 99

Reviewer:

Air Force - 85

Preparing Activity:

Air Force - 84

Project Number:

3950-F005

☆ U.S. GOVERNMENT PRINTING OFFICE: 1988—605-036/21703

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:19Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-W-26217 B	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:19Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

☆ U.S. Government Printing Office: 1982—360-979/5804

(Fold along this line)

(Fold along this line)

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 73236 WASH DC

POSTAGE WILL BE PAID BY HQ U.S. AIR FORCE

WR-ALC/MMEDTA
ROBINS AFB, GA. 31098-5609

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:19Z
Check the source to verify that this is the current version before use.